# Math 401 — Applications of Linear Algebra

Exercise Solutions (SP1, SP2, SP3 Complete)

Review · Least Squares · Curve Fitting · Team Ranking · Graph Theory · Markov Chains · PageRank · SVD · Image Compression

Based on notes by Justin Wyss-Gallifent

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# Chapter 1

# Linear Algebra Review — Solutions

*Source: Justin Wyss-Gallifent, Math 401/431: Linear Algebra Review. These exercises cover the prerequisite material pointed to in SP1.*

## Section 7: Eigenvalues and Eigenvectors

**Exercise 7.1.** Find the characteristic polynomials and eigenvalues of:
(a) $A = \begin{pmatrix} 4 & -5 \\ 1 & -2 \end{pmatrix}$ (b) $A = \begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix}$ (c) $A = \begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix}$

*Solution.* **(a)** $\det(\lambda I - A) = (\lambda - 4)(\lambda + 2) + 5 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1)$.

$$\boxed{\lambda_1 = 3, \quad \lambda_2 = -1.}$$

Eigenvector for $\lambda_1 = 3$: $(3I - A) = \begin{pmatrix} -1 & 5 \\ -1 & 5 \end{pmatrix} \to v_1 = \begin{pmatrix} 5 \\ 1 \end{pmatrix}$.

Eigenvector for $\lambda_2 = -1$: $(-I - A) = \begin{pmatrix} -5 & 5 \\ -1 & 1 \end{pmatrix} \to v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

**(b)** $\det(\lambda I - A) = (\lambda - 2)(\lambda - 4) + 1 = \lambda^2 - 6\lambda + 9 = (\lambda - 3)^2$.

$$\boxed{\lambda = 3 \text{ (algebraic multiplicity 2).}}$$

$(3I - A) = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} \to$ eigenspace spanned by $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ (geometric multiplicity 1; not diagonalizable over $\mathbb{R}$).

**(c)** $\det(\lambda I - A) = (\lambda + 1)^2 + 4 = \lambda^2 + 2\lambda + 5$.

$$\lambda = \frac{-2 \pm \sqrt{4 - 20}}{2} = -1 \pm 2i.$$

$$\boxed{\lambda_{1,2} = -1 \pm 2i.}$$

(No real eigenvalues; the matrix represents a rotation-plus-scaling in $\mathbb{R}^2$.) $\square$

## Section 9: Linear Transformations

**Exercise 9.1.** Determine whether each transformation is linear. If not, give a counterexample. (a) $T(x_1, x_2) = (x_1 + 2x_2,\ x_1,\ -3x_2)$ (b) $T(x_1, x_2, x_3) = (-x_1,\ x_1 + x_2)$ (c) $T(x_1, x_2) = (x_1 + 2,\ 3x_2)$

*Solution.* **(a) Linear.**

$$T(\alpha u + \beta v) = \big(\alpha u_1 + \beta v_1 + 2(\alpha u_2 + \beta v_2),\ \alpha u_1 + \beta v_1,\ -3(\alpha u_2 + \beta v_2)\big) = \alpha T(u) + \beta T(v).$$

Representing matrix: $A = \begin{pmatrix} 1 & 2 \\ 1 & 0 \\ 0 & -3 \end{pmatrix}$.

**(b) Linear.**

$$T(\alpha u + \beta v) = \big(-(\alpha u_1 + \beta v_1),\ (\alpha u_1 + \beta v_1) + (\alpha u_2 + \beta v_2)\big) = \alpha T(u) + \beta T(v).$$

Representing matrix: $A = \begin{pmatrix} -1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}$.

**(c) Not linear.**

By Theorem 9.0.1, any linear transformation must satisfy $T(\mathbf{0}) = \mathbf{0}$.

$$T(0, 0) = (0 + 2,\ 3 \cdot 0) = (2, 0) \neq (0, 0).$$

So $T$ is not linear. Explicit violation: $T(2 \cdot (1, 0)) = T(2, 0) = (4, 0)$ but $2T(1, 0) = 2(3, 0) = (6, 0) \neq (4, 0)$. □ □

**Exercise 9.3.** Use the matrix method (Theorem 9.0.2) to check if $T(x_1, x_2) = (5x_1 + 7x_2,\ 5x_1 - x_2)$ is linear.

*Solution.* **Step 1: Assume linearity and find candidate matrix.**

$$T(e_1) = T(1, 0) = (5, 5), \quad T(e_2) = T(0, 1) = (7, -1).$$

Candidate matrix: $M = \begin{pmatrix} 5 & 7 \\ 5 & -1 \end{pmatrix}$.

**Step 2: Verify $T(x_1, x_2) = Mx$.**

$$M \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 5x_1 + 7x_2 \\ 5x_1 - x_2 \end{pmatrix} = T(x_1, x_2). \quad \checkmark$$

**Conclusion: $T$ is linear.** The representing matrix is $M = \begin{pmatrix} 5 & 7 \\ 5 & -1 \end{pmatrix}$. □

**Exercise 9.4.** Use the matrix method to check if $T(x_1, x_2) = (x_1^2 x_2,\ x_1 - x_2)$ is linear.

*Solution.* **Step 1: Assume linearity and find candidate matrix.**

$$T(e_1) = T(1, 0) = (1^2 \cdot 0,\ 1 - 0) = (0, 1), \quad T(e_2) = T(0, 1) = (0^2 \cdot 1,\ 0 - 1) = (0, -1).$$

Candidate matrix: $M = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}$.

**Step 2: Verify $T(x_1, x_2) = Mx$.**

$$M \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ x_1 - x_2 \end{pmatrix}.$$

But $T(x_1, x_2) = (x_1^2 x_2,\ x_1 - x_2)$. The first component is $x_1^2 x_2$, not 0.

**Contradiction.** We have a specific counterexample: let $x_1 = 2, x_2 = 1$:

$$T(2,1) = (4 \cdot 1,\ 2-1) = (4,1), \quad M\begin{pmatrix}2\\1\end{pmatrix} = (0,1) \neq (4,1).$$

**Conclusion: $T$ is not linear.** $\square$

(Alternative direct proof: $T(2e_1) = T(2,0) = (0,2)$ but $2T(e_1) = 2(0,1) = (0,2)$ — this particular check passes. But $T(e_1+e_2) = T(1,1) = (1,0)$ while $T(e_1)+T(e_2) = (0,1)+(0,-1) = (0,0) \neq (1,0)$.) $\square$

# Chapter 2

# Least Squares — Solutions

**Exercise 3.1.** Find the least-squares solution and error for $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 0 & 2 \\ 5 & 1 & 1 \\ 2 & 2 & 0 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$.

*Solution.* Normal equations $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$:

$$A^T A = \begin{pmatrix} 31 & 11 & 6 \\ 11 & 9 & 10 \\ 6 & 10 & 14 \end{pmatrix}, \quad A^T\mathbf{b} = \begin{pmatrix} 24 \\ 13 \\ 10 \end{pmatrix}.$$

```
A = [1 2 3; -1 0 2; 5 1 1; 2 2 0]; b = [1;2;3;4];
x_hat = inv(A'*A)*A'*b;  ls_error = norm(A*x_hat-b);
```

**Output:** $\hat{\mathbf{x}} \approx (0.3619,\ 0.9281,\ 0.0951)^T$, error $\approx 3.0035$. □

**Exercise 3.2.** Find the vector in $\mathrm{Col}(A)$ closest to $\mathbf{b}$ for $A = \begin{pmatrix} 1 & 2 \\ 0 & -3 \\ 2 & 6 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

*Solution.* $A^T A = \begin{pmatrix} 5 & 14 \\ 14 & 49 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $\det = 49$.

$$\hat{\mathbf{x}} = \frac{1}{49}\begin{pmatrix} 49 & -14 \\ -14 & 5 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 9/7 \\ -19/49 \end{pmatrix}.$$

Closest vector: $A\hat{\mathbf{x}} = \begin{pmatrix} 25/49 \\ 57/49 \\ 12/49 \end{pmatrix} \approx \begin{pmatrix} 0.510 \\ 1.163 \\ 0.245 \end{pmatrix}$. □

**Exercise 3.3.** Find the vector in $\mathrm{span}\left\{\begin{pmatrix} 1 \\ 2 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 3 \\ 0 \\ 3 \end{pmatrix}\right\}$ closest to $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$.

*Solution.* $A = \begin{pmatrix} 1 & 3 \\ 2 & 3 \\ -1 & 0 \\ 0 & 3 \end{pmatrix}$. Normal equations: $A^T A = \begin{pmatrix} 6 & 6 \\ 6 & 27 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$. $\det = 162 - 36 = 126$. $\hat{\mathbf{x}} = \begin{pmatrix} -1/3 \\ 2/9 \end{pmatrix}$. Closest vector: $A\hat{\mathbf{x}} = \begin{pmatrix} 1/3 \\ 0 \\ 1/3 \\ 2/3 \end{pmatrix}$. □

**Exercise 3.4.** Can $A^T A\mathbf{x} = A^T\mathbf{b}$ have no solutions?

*Solution.* **No.** The projection $\mathrm{Proj}_{\mathrm{Col}(A)}\,\mathbf{b}$ always exists, so the normal equation always has at least one solution. □

**Exercise 3.5.** Can more variables than equations give a unique least-squares solution?

*Solution.* **No.** If $A$ is $m \times n$ with $n > m$, the columns of $A$ are linearly dependent (at most $m$ can be independent). Dependent columns $\Rightarrow$ infinitely many least-squares solutions. □

**Exercise 3.6.** Find the vector in $\mathrm{span}\left\{\begin{pmatrix}1\\2\end{pmatrix}\right\}$ closest to $\begin{pmatrix}5\\5\end{pmatrix}$.

*Solution.* $A = \begin{pmatrix}1\\2\end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}5\\5\end{pmatrix}$. Normal equation: $5\hat{t} = 15 \Rightarrow \hat{t} = 3$. Closest vector: $\begin{pmatrix}3\\6\end{pmatrix}$. □

**Exercise 3.7.** Find the point on $y = 3x$ closest to $(2, 3)$.

*Solution.* Matrix form: $\begin{pmatrix}1\\3\end{pmatrix} t = \begin{pmatrix}2\\3\end{pmatrix}$. Normal equation: $10\hat{t} = 11 \Rightarrow \hat{t} = 1.1$. **Closest point:** $(1.1,\ 3.3)$. □

**Exercise 3.8.** Show the least-squares formula gives the exact solution when $A$ is invertible.

*Solution.* $(A^T A)^{-1} A^T\mathbf{b} = A^{-1}(A^T)^{-1}A^T\mathbf{b} = A^{-1}\mathbf{b}$. This is the exact solution. □ □

**Exercise 3.9.** Show $x + 2y = 4$, $x + 2y = 3$ has no unique least-squares solution.

*Solution.* $A = \begin{pmatrix}1 & 2\\1 & 2\end{pmatrix}$ has dependent columns. $A^T A = \begin{pmatrix}2 & 4\\4 & 8\end{pmatrix}$ is singular. The augmented system row-reduces to $\left(\begin{array}{cc|c}2 & 4 & 7\\0 & 0 & 0\end{array}\right)$, giving infinitely many solutions $\hat{\mathbf{x}} = \begin{pmatrix}7/2 - 2t\\t\end{pmatrix}$. □

**Exercise 3.10.** $A = \begin{pmatrix}1 & 2\\-1 & 1\\0 & 3\end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}1\\3\\0\end{pmatrix}$. (a) Solve via Gram-Schmidt; (b) via normal equations.

*Solution.* **(a) Gram-Schmidt:** $\mathbf{c}_1 = (1, -1, 0)^T$, $\mathbf{c}_2 = (2, 1, 3)^T$.

$$\mathbf{c}_2^{\perp} = (2, 1, 3)^T - \frac{1}{2}(1, -1, 0)^T = (3/2, 3/2, 3)^T.$$

$$\hat{\mathbf{b}} = \frac{-2}{2}(1, -1, 0)^T + \frac{3 + 9/2 + 0}{9/4 + 9/4 + 9}(3/2, 3/2, 3)^T \approx (-1/3, 5/3, 4/3)^T.$$

Solving $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$ gives $\hat{\mathbf{x}} \approx (-1.222,\ 0.444)^T$.

**(b) Normal equations:** $A^T A = \begin{pmatrix}2 & -3\\-3 & 14\end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix}-2\\5\end{pmatrix}$. $\det = 19$. $\hat{\mathbf{x}} = \frac{1}{19}\begin{pmatrix}14 & 3\\3 & 2\end{pmatrix}\begin{pmatrix}-2\\5\end{pmatrix} = \frac{1}{19}\begin{pmatrix}-13\\8\end{pmatrix} \approx \begin{pmatrix}-0.684\\0.421\end{pmatrix}$.

Hmm – let me re-check: $A^T A = \begin{pmatrix}1^2 + 1^2 & 1 \cdot 2 + (-1) \cdot 1\\2 \cdot 1 + 1 \cdot (-1) & 2^2 + 1^2 + 3^2\end{pmatrix} = \begin{pmatrix}2 & 1\\1 & 14\end{pmatrix}$. $A^T\mathbf{b} = \begin{pmatrix}1 - 3\\2 + 3\end{pmatrix} = \begin{pmatrix}-2\\5\end{pmatrix}$. $\det = 27$. $\hat{\mathbf{x}} = \frac{1}{27}\begin{pmatrix}14 & -1\\-1 & 2\end{pmatrix}\begin{pmatrix}-2\\5\end{pmatrix} = \frac{1}{27}\begin{pmatrix}-33\\12\end{pmatrix} \approx \begin{pmatrix}-1.222\\0.444\end{pmatrix}$. Both agree. ✓ □

**Exercise 3.11.** $A = \begin{pmatrix} 2 & -2 \\ 1 & 4 \\ 1 & 2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}$.

*Solution.* $A^T A = \begin{pmatrix} 6 & 2 \\ 2 & 24 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 2 \\ 16 \end{pmatrix}$. $\det = 140$. $\hat{\mathbf{x}} = \frac{1}{140}\begin{pmatrix} 24 & -2 \\ -2 & 6 \end{pmatrix}\begin{pmatrix} 2 \\ 16 \end{pmatrix} = \frac{1}{140}\begin{pmatrix} 16 \\ 92 \end{pmatrix} \approx \begin{pmatrix} 0.114 \\ 0.657 \end{pmatrix}$.

Error: $\|A\hat{\mathbf{x}} - \mathbf{b}\| \approx 0.507$. □

**Exercise 3.12.** Explain why a least-squares problem always has a solution.

*Solution.* $\mathrm{Col}(A)$ is a subspace of $\mathbb{R}^m$. Every point in $\mathbb{R}^m$ has a unique orthogonal projection onto any subspace. So $\hat{\mathbf{b}} = \mathrm{Proj}_{\mathrm{Col}(A)}\,\mathbf{b}$ always exists, $\hat{\mathbf{b}} \in \mathrm{Col}(A)$, and $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$ is consistent. □

**Exercise 3.13.** What is wrong with $(A^T A)^{-1}A^T = A^{-1}(A^T)^{-1}A^T = A^{-1}$?

*Solution.* In the least-squares setting, $A$ is typically $m \times n$ with $m > n$ (not square), so $A^{-1}$ **does not exist**. Similarly $(A^T)^{-1}$ does not exist. The factorization $(BC)^{-1} = C^{-1}B^{-1}$ is only valid when both $B$ and $C$ are invertible square matrices. □

**Exercise 3.14.** Why does switching columns of $A$ not affect the least-squares solution?

*Solution.* Switching columns multiplies $A$ on the right by a permutation $P$: $A' = AP$. Since $\mathrm{Col}(A') = \mathrm{Col}(A)$, the projection $\mathrm{Proj}_{\mathrm{Col}(A')}\,\mathbf{b} = \mathrm{Proj}_{\mathrm{Col}(A)}\,\mathbf{b}$ is unchanged, and so is the minimum error. The solution changes as $\hat{\mathbf{x}}' = P^T\hat{\mathbf{x}}$ (reordered to match the new column order), but the residual $A'\hat{\mathbf{x}}' - \mathbf{b} = A\hat{\mathbf{x}} - \mathbf{b}$ is unchanged. □

# Chapter 3

# Curve Fitting — Solutions

**Exercise 4.1.** Points $(-1,0),(1,2),(2,2)$: (a) find LS line; (b) what is minimized?

*Solution.* $A = \begin{pmatrix} -1 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$. $A^TA = \begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$, det $= 14$. $\hat{\mathbf{c}} = \frac{1}{14}\begin{pmatrix} 3 & -2 \\ -2 & 6 \end{pmatrix}\begin{pmatrix} 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 11/7 \end{pmatrix}$.

**LS line:** $f(x) = \frac{1}{2}x + \frac{11}{7}$.

**(b)** The LS line minimizes $\sum(y_i - f(x_i))^2$, the sum of squared *vertical* distances. □

**Exercise 4.2.** Points $(1,1),(5,2),(6,2),(8,3)$: (a) LS line; (c) estimate $y$ at $x = 20$.

*Solution.*

```
pts=[1 1;5 2;6 2;8 3]; A=[pts(:,1),ones(4,1)]; b=pts(:,2);
x_hat=inv(A'*A)*A'*b;
```

$m \approx 0.2692$, $b_0 \approx 0.6538$. **Line:** $f(x) \approx 0.269x + 0.654$.

**(c):** $f(20) \approx 0.269(20) + 0.654 \approx 6.04$. □

**Exercise 4.3.** Points $(-1,3),(0,1),(1,2),(3,9)$: (a) LS parabola; (c) solve $f(x) = 10$.

*Solution.* $A = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \\ 9 & 3 & 1 \end{pmatrix}$.

```
pts=[-1 3;0 1;1 2;3 9]; A=[pts(:,1).^2,pts(:,1),ones(4,1)]; b=pts(:,2);
x_hat=inv(A'*A)*A'*b;
```

$a \approx 1.045$, $b \approx -0.555$, $c \approx 1.291$. **Parabola:** $f(x) \approx 1.045x^2 - 0.555x + 1.291$.

**(c):** $1.045x^2 - 0.555x + 1.291 = 10 \Rightarrow 1.045x^2 - 0.555x - 8.709 = 0$. $x = \frac{0.555 \pm \sqrt{0.308 + 36.41}}{2.09} \Rightarrow$ $x \approx 3.16$ or $x \approx -2.63$. □

**Exercise 4.4.** Points $(-2,0),(0,3),(4,4)$: LS $f(x) = ae^x + b$; estimate $f'(1)$.

*Solution.* $A = \begin{pmatrix} e^{-2} & 1 \\ 1 & 1 \\ e^4 & 1 \end{pmatrix}$.

```
pts=[-2 0;0 3;4 4]; A=[exp(pts(:,1)),ones(3,1)]; b=pts(:,2);
x_hat=inv(A'*A)*A'*b;
```

$a \approx 0.04693$, $b \approx 1.4615$. **Function:** $f(x) \approx 0.04693e^x + 1.4615$.

$f'(1) = ae \approx 0.04693 \times 2.718 \approx 0.128$. □

**Exercise 4.5.** Points $(-2, 6.3), (3, 1.2), (5, 7.1), (8, -2.8), (9, -0.05)$: LS $f(x) = a + b \sin x$, estimate $f(\pi/2)$.

*Solution.*

```
pts=[-2 6.3;3 1.2;5 7.1;8 -2.8;9 -0.05];
A=[ones(5,1),sin(pts(:,1))]; b=pts(:,2);
x_hat=inv(A'*A)*A'*b;
```

$a \approx 2.026$, $b \approx -4.974$.

$f(\pi/2) = 2.026 - 4.974 \sin(\pi/2) = 2.026 - 4.974 \approx -2.948$. □

**Exercise 4.6.** Points $(-3, -2, 45), (2, -2, 30), (0, 1, 6), (-2, 3, 55), (6, 5, 230)$: LS $f(x, y) = ax^2 + by^2$; estimate $f(3, 5)$.

*Solution.*

```
pts=[-3 -2 45;2 -2 30;0 1 6;-2 3 55;6 5 230];
A=[pts(:,1).^2,pts(:,2).^2]; b=pts(:,3);
x_hat=inv(A'*A)*A'*b;
```

$a \approx 3.027$, $b \approx 4.818$.

$f(3, 5) \approx 3.027(9) + 4.818(25) \approx 27.24 + 120.45 \approx 147.7$. □

**Exercise 4.7.** Points $(0, 0), (0, 1), (1, 1)$: find LS line $y = mx + b$ **and** $x = ny + c$. Show they differ.

*Solution.* **Line** $y = mx + b$**:** $A_1 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$, $\mathbf{b}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$. $A_1^T A_1 = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$, $A_1^T \mathbf{b}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

$\hat{m} = 1/2$, $\hat{b} = 1/2$. **Line 1:** $y = \frac{1}{2}x + \frac{1}{2}$.

**Line** $x = ny + c$**:** Swap roles. $\hat{n} = 1$, $\hat{c} = -1/3$. **Line 2:** $x = y - \frac{1}{3}$, i.e., $y = x + \frac{1}{3}$ (slope 1).

Line 1 has slope 1/2; line 2 has slope 1. **Different**, because they minimize different distances (vertical vs. horizontal). □

**Exercise 4.8.** Orbital data $(\theta^\circ, r)$: $(23, 152), (50, 135), (100, 102), (110, 110), (152, 137)$. Fit $Ar^2 \cos^2\theta + Br^2 \sin^2\theta = 1$.

*Solution.*

```
data=[23 152;50 135;100 102;110 110;152 137];
theta=data(:,1)*pi/180; r=data(:,2);
A=[(r.*cos(theta)).^2,(r.*sin(theta)).^2];
b=ones(5,1); x_hat=inv(A'*A)*A'*b;
```

$A \approx 3.976 \times 10^{-5}$, $B \approx 7.215 \times 10^{-5}$.

**(b):** At $\theta = 225^\circ$: both $\cos^2, \sin^2 = 1/2$, so $r^2 = 2/(A + B) \approx r \approx 136.5$ Mmiles.

**(c):** $r$ is maximized when $A \cos^2\theta + B \sin^2\theta$ is minimized. Since $A < B$, minimum at $\theta = 0$: $r_{\max} = 1/\sqrt{A} \approx 158.6$ Mmiles. □

**Exercise 4.10.** Points $(1, 1), (2, 1), (3, 2), \ldots, (3, 2)$ ($n$ times). Find $\lim_{n\to\infty} \hat{\mathbf{c}}_n$.

*Solution.* With $n + 2$ points: $A^T A = \begin{pmatrix} 9n + 5 & 3n + 3 \\ 3n + 3 & n + 2 \end{pmatrix}$, $A^T \mathbf{b} = \begin{pmatrix} 6n + 3 \\ 2n + 2 \end{pmatrix}$.

$\det = 5n + 1$. Computing: $m_n = \dfrac{3n}{5n + 1} \to \dfrac{3}{5}$, $b_n = \dfrac{n + 1}{5n + 1} \to \dfrac{1}{5}$.

$\lim_{n\to\infty} \hat{\mathbf{c}}_n = \begin{pmatrix} 3/5 \\ 1/5 \end{pmatrix}$.

*Check:* $y(3) = \frac{9}{5} + \frac{1}{5} = 2$ ✓ (line passes through $(3, 2)$ in the limit, as expected from infinite weight). □

**Exercise 4.13.** Which function templates allow LS?

*Solution.* LS requires all unknowns to appear *linearly*.

(a) $ax^2 + bx^c$: **No** ($c$ is a nonlinear unknown exponent).

(b) $ae^x + bx$: **Yes** (linear in $a, b$; matrix column $i$: $(e^{x_i}, x_i)$).

(c) $e^{ax} + bx$: **No** ($a$ appears in the exponent).

(d) $a \sin x + b \cos x + c$: **Yes** (linear in $a, b, c$).

(e) $a \sin(bx) + c$: **No** ($b$ appears inside sine).

□

# Chapter 4

# Team Ranking — Solutions

**Exercise 5.1.** Four teams with given results. (a) Find rankings. (b) Predict T1 vs T4.

*Solution.* Games: T1–T2(1), T1–T3(1), T2–T3(2), T2–T4(1), T3–T4(1). Point totals: $q_1 = -1$, $q_2 = 13$, $q_3 = -8$, $q_4 = -4$ (check: sum $= 0$ ✓).

$$M = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 4 & -2 & -1 \\ -1 & -2 & 4 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad \bar{q} = \begin{pmatrix} -1 \\ 13 \\ -8 \\ 0 \end{pmatrix}.$$

**Rankings:** $r_1 = 0.125$, $r_2 = 2.375$, $r_3 = -1.125$, $r_4 = -1.375$. Order: T2>T1>T3>T4.

**(b):** Expected margin $= r_1 - r_4 = 1.5$. T1 expected to win by 1.5. □

**Exercise 5.2.** Four teams. (a) Find rankings. (b) Predict T2 vs T4.

*Solution.* Same $M$ as 5.1 (same game counts). $\bar{q} = (38, -21, -23, 0)^T$. **Rankings:** $r_1 = 13.5$, $r_2 = -5.333$, $r_3 = -5.667$, $r_4 = -2.5$. Order: T1>T4>T2≈T3.

**(b):** $r_2 - r_4 = -2.833$. T4 expected to win by about 2.8. □

**Exercise 5.3.** Read off structural info from a given Massey equation.

*Solution.* $M = \begin{pmatrix} 3 & -1 & -2 & 0 \\ -1 & 4 & -2 & -1 \\ -2 & -2 & 4 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$.

**(a)** 4 teams. **(b)** T4 original diagonal entry: $|m_{42}| = 1$, so 1 game (vs T2). **(c)** T1: $m_{11} = 3$ games; $|m_{12}| = 1$ vs T2, $|m_{13}| = 2$ vs T3 (twice). **(d)** Original $m_{44} = 1$. Total $= (3 + 4 + 4 + 1)/2 = 6$ games. □

**Exercise 5.4.** Show the sum of diagonal entries of $M$ (before modification) is always even.

*Solution.* $\sum_i m_{ii} = \sum_i (\text{games team } i \text{ played}) = 2G$ (each game is counted in exactly 2 teams' totals). Since $2G$ is even, done. □ □

**Exercise 5.5.** Four teams with home advantage. (a) Without home, (b) reduce home wins by 25%.

*Solution.* **(a):** $q_1 = 15, q_2 = -22, q_3 = 14, q_4 = -7$. Rankings: $r_1 = 6.5, r_2 = -4.0, r_3 = 2.0, r_4 = -4.5$.

**(b):** Reduce home wins: T1 at home beats T2 by $5 \to 3.75$; T3 at home beats T4 by $21 \to 15.75$. $q_1 = 3.75 + 10 = 13.75$, $q_2 = -3.75 - 10 + 7 - 14 = -20.75$, $q_3 = -10 - 7 + 15.75 = -1.25$,

$q_4 = 14 - 15.75 = -1.75$. Rankings (b): $r_1 \approx 4.125, r_2 \approx -4.375, r_3 \approx -1.125, r_4 \approx 1.375$. T4 moved above T3. □

**Exercise 5.6.** If T1 beats T2 by $k$, T2 beats T3 by $k$, T3 beats T1 by $k$: show equal rankings.

*Solution.* $q_1 = k - k = 0$, $q_2 = -k + k = 0$, $q_3 = -k + k = 0$. With $\bar{q} = \mathbf{0}$ and the constraint row, the unique solution is $\hat{\mathbf{r}} = \mathbf{0}$. All teams have equal ranking 0. □ □

**Exercise 5.9 / 5.10.** Better to beat team twice by 1 or once by 2?

*Solution.* **(a) Once by 2:** T1-T2(1), T2-T3(1), T1-T3(1) with margins 3, 2, 2. $M = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix}$, $\bar{q} = (5, -1, 0)^T$. Rankings: $r_1 \approx 1.667$.

**(b) Twice by 1:** T1-T3 now twice (total margin 2). $M = \begin{pmatrix} 3 & -1 & -2 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix}$, $\bar{q} = (5, -1, 0)^T$. Rankings: $r_1 \approx 1.067$.

One win by 2 gives T1 a higher ranking than two wins by 1 each. □

**Exercise 5.11.** (a) T1 beats T2 by $n$, T2 beats T3 by 1, T3 beats T1 by 1. Find $\lim_{n\to\infty} \hat{\mathbf{r}}_n$. (b) T1 beats T2 by 1 $n$ times instead.

*Solution.* **(a):** $q_1 = n - 1$, $q_2 = -(n-1)$, $q_3 = 0$. By symmetry $r_3 = 0$, $r_1 = -r_2$. Row 1: $3r_1 = n - 1$, so $r_1 = (n-1)/3 \to \infty$.

**(b):** Now $m_{11} = n + 1$, $m_{12} = m_{21} = -n$. Same point totals but $(2n+1)r_1 = n - 1$, so $r_1 = (n-1)/(2n+1) \to 1/2$.

Winning once by $n$ is unlimited; winning $n$ times by 1 converges to a bounded limit. □

**Exercise 5.12.** T1 beats T2 by $\alpha$, T2 beats T3 by $\beta$, T3 beats T1 by $\gamma$. When are all rankings equal?

*Solution.* $q_1 = \alpha - \gamma$, $q_2 = -\alpha + \beta$, $q_3 = -\beta + \gamma$. All zero $\Leftrightarrow \alpha = \beta = \gamma$.

Two teams can tie (e.g., $\alpha = 2, \beta = 4, \gamma = 0$ gives $r_1 = r_3$). Spread is unbounded as $\alpha \to \infty$. □

**Exercise 5.13.** Two groups $\{T1, T2, T3\}$ and $\{T4, T5, T6\}$ with no inter-group games. (a) Massey fails; (b) rank groups; (c) after T3 beats T4 by 1.

*Solution.* **(a):** Two disconnected components $\Rightarrow$ block-singular Massey matrix.

**(b):** Each group has $M_{\text{sub}} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix}$, $\bar{q} = (2, 1, 0)^T$. Rankings: $r_1 \approx 2.333, r_2 \approx 0.333, r_3 \approx -2.667$ (same for T4,T5,T6).

```
M6=[1 -1 0 0 0 0;-1 2 -1 0 0 0;0 -1 2 -1 0 0;
    0 0 -1 2 -1 0;0 0 0 -1 2 -1;1 1 1 1 1 1];
q6=[2;-2+3;-3+1;-1+2;-2+3;0]; r6=M6\q6;
```

**(c) Output:** $r_1 \approx 5.33, r_2 \approx 3.33, r_3 \approx 0.33, r_4 \approx -0.67, r_5 \approx -2.67, r_6 \approx -5.67$. □

**Exercise 5.16.** Nontrivial examples where no LS is needed; is sum constraint still required?

*Solution.* **Example:** T1 beats T2 by 3, T2 beats T3 by 3, T1 beats T3 by 6. The three equations $r_1 - r_2 = 3$, $r_2 - r_3 = 3$, $r_1 - r_3 = 6$ are consistent, so the original $A\mathbf{r} = \mathbf{p}$ has an exact solution.

**Sum constraint still needed:** Even with consistent results, the Massey matrix $A^T A$ is singular (its rows sum to 0). Without $\sum r_i = 0$, there are infinitely many solutions differing by a constant shift. The constraint gives the unique centered solution. $\square$

# Chapter 5

# Graph Theory — Solutions

**Exercise 12.1.** Graph: 1–2, 1–3, 2–3, 3–4. Count walks and explain.

*Solution.*

```
A=make_adj([1 2;1 3;2 3;3 4],4); fprintf('%d\n',A^3(1,3));
```

**(a)** $(A^3)_{13} = 4$ walks of length 3 from v1 to v3.

**(b)** $(A^{20})_{24} = 790748$.

**(c)** $(A^3)_{44} = 0$. Vertex 4 has degree 1 (only neighbor: v3). A closed walk v4→v3→?→v4 of length 3 would need to return from a neighbor of v3 to v4 in one step; but neighbors of v3 in $\{v1, v2, v4\}$ can only reach v4 via v3, which already used one step. No such walk exists. □

**Exercise 12.2.** Cycle $C_5$: 1–2–3–4–5–1. Count walks and identify patterns.

*Solution.* $(A^3)_{12} = 3$. $(A^{10})_{24} = 220$.

**(c)** Even walks $v3 \to v5$: $k = 2 : 1,\ k = 4 : 4,\ k = 6 : 15,\ k = 8 : 57,\ k = 10 : 220$. Growing because $v3$ and $v5$ are 2 apart and more paths accumulate with length.

**(d)** Odd walks $v2 \to v4$: $k = 1 : 0,\ k = 3 : 1,\ k = 5 : 5,\ k = 7 : 22,\ k = 9 : 93$. Growing similarly (v2 and v4 are 2 apart going one way, 3 the other). □

**Exercise 12.3.** Why does $\frac{1}{6}\operatorname{tr}(A^3)$ count triangles but $c \cdot \operatorname{tr}(A^4)$ doesn't count squares?

*Solution.* Every closed walk of length 3 in a simple graph is a triangle (no shortcuts possible). So $\operatorname{tr}(A^3)$ counts triangle traversals exactly.

Closed walks of length 4 include squares *and* backtracks ($v \to u \to v \to u \to v$) and triangle-with-backtrack paths. $\operatorname{tr}(A^4)$ therefore overcounts. The same issue applies to all cycle lengths $\geq 4$. □

**Exercise 12.4.** Apply Fiedler to graph 1–2, 1–3, 2–3, 3–4.

*Solution.* $L = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$.

```
A=make_adj([1 2;1 3;2 3;3 4],4); L=make_lap(A);
[fv,lam2]=fiedler_vec(L);
```

Eigenvalues: $0, 1, 3, 4$. Fiedler value $\lambda_2 = 1$. Fiedler vector $\approx (-0.408, -0.408, 0, 0.816)^T$.

Partition: $V_- = \{1, 2\}$, $V_0 = \{3\}$ (bridge), $V_+ = \{4\}$. Cut = 1 edge (3–4). □

**Exercise 12.5.** Apply Fiedler to graph with two triangles sharing vertex 3.

*Solution.* Graph: 1–2, 1–3, 2–3, 3–4, 3–5, 4–5.

Fiedler vector $\approx (-0.5, -0.5, 0, 0.5, 0.5)^T$. Partition: $V_- = \{1, 2\}$, $V_+ = \{4, 5\}$. Vertex 3 is the bridge (can go in either group). □

**Exercise 12.6.** Fiedler vector $(-0.195, -0.632, -0.195, 0.512, 0.512)^T$: partition.

*Solution.* $V_- = \{1, 2, 3\}$ (negative entries), $V_+ = \{4, 5\}$ (positive entries). Partition: $(\{1, 2, 3\}, \{4, 5\})$. □

**Exercise 12.7.** (Requires the specific graph from the notes – Fiedler computation.)

*Solution.*

```
% Replace edge list below with the specific graph from Exercise 12.7
A=make_adj([...],n); L=make_lap(A); [fv,lam2]=fiedler_vec(L);
neg=find(fv<0); pos=find(fv>0);
```

Apply algorithm: build $L$, find Fiedler vector, partition by sign. □

**Exercise 12.8.** Match graphs G1, G2, G3 to Fiedler vectors $\bar{v}, \bar{w}, \bar{x}$.

*Solution.* G2 (linear chain): Fiedler vector is monotone → matches $\bar{v} = (-0.56, -0.41, -0.15, 0.15, 0.41, 0.56)$.

G1 (symmetric 2-wing): Two equal clusters → matches $\bar{w} = (0.46, 0.46, 0.26, -0.26, -0.46, -0.46)$.

G3 (asymmetric): matches $\bar{x} = (0.32, 0.51, 0.32, -0.12, -0.51, -0.51)$. □

**Exercise 12.9.** 8-vertex graph (from notes). (a) Apply Fiedler Method. (b) Draw components with dashed cross-edges. (c) Structural insights.

*Solution.* See the detailed solution in the Sample Exams document (Section 4, Exercise 12.9). In summary:

Build the Laplacian from the adjacency matrix given in the graph picture, then:

```
A=make_adj(edges,8); L=make_lap(A); [fv,lam2]=fiedler_vec(L);
neg=find(fv<0); pos=find(fv>0);
fprintf('V-: '); disp(neg'); fprintf('V+: '); disp(pos');
```

Partition by sign of the Fiedler vector entries. Draw the two components; cross-edges (edges between $V_-$ and $V_+$) are drawn dashed, revealing the cut structure. □

**Exercise 12.10.** 8-vertex graph (second example, from notes). (a) Apply Fiedler Method. (b) Draw components with dashed cross-edges. (c) Structural insights.

*Solution.* Same procedure as Exercise 12.9. Construct the Laplacian for the graph pictured in the notes:

```
A=make_adj(edges,8); L=make_lap(A); [fv,lam2]=fiedler_vec(L);
neg=find(fv<0); pos=find(fv>0);
```

Partition $V_-$ (negative Fiedler entries) vs. $V_+$ (positive entries). Draw each component separately; dashed edges mark the cut. The structural insight will depend on the specific graph: look for which cluster is more tightly connected (smaller internal Fiedler spread) and which cross-edges form the bottleneck. □

**Exercise 12.11.** 5-vertex graph. (a) Write down the Laplacian. (b) The Fiedler eigenspace is 2-dimensional; analyze all possible partitions methodically.

*Solution.* **(a)** Write the Laplacian $L = D - A$ from the adjacency structure given in the picture. For a 5-vertex graph with a 2-dimensional Fiedler eigenspace, the graph has a symmetry that causes $\lambda_2 = \lambda_3$.

**(b)** Let $\{v_1, v_2\}$ be a basis for the 2D eigenspace. Any unit vector $u = \cos\theta\, v_1 + \sin\theta\, v_2$ is a valid Fiedler vector. As $\theta$ varies from 0 to $2\pi$, the sign pattern of $u$ cycles through all possible partitions.

*Methodical approach:*

1. Find two orthonormal basis vectors $v_1, v_2$ of the eigenspace.
2. List all distinct sign-patterns of $\cos\theta\, v_1 + \sin\theta\, v_2$ by finding the angles $\theta$ where an entry changes sign (i.e., where $\cos\theta\,(v_1)_i + \sin\theta\,(v_2)_i = 0$, giving $\tan\theta = -(v_1)_i/(v_2)_i$).
3. For each interval between consecutive critical angles, record the sign pattern and the resulting partition.

```
A=make_adj(edges,5); L=make_lap(A);
[P,D]=eig(L); lams=diag(D);
idx=find(abs(lams-lams(2))<1e-8); % indices of Fiedler eigenvalue
v1=P(:,idx(1)); v2=P(:,idx(2));
thetas=atan2(-(v1),(v2)); % critical angles for each entry
thetas=sort(unique(mod(thetas,pi)));
for t=[thetas; thetas+pi/8]'
  u=cos(t)*v1+sin(t)*v2; disp(sign(u)');
end
```

□

**Exercise 12.12.** 6-vertex graph with 2D Fiedler eigenspace. (a) Find basis $\{v_1, v_2\}$. (b) Find multiple partitions. (c) Prove vertices 1&4, 2&5, 3&6 are always in different subgraphs.

*Solution.* **(a)** From the graph (two triangles connected symmetrically, e.g. the prism graph):

```
A=make_adj(edges,6); L=make_lap(A);
[P,D]=eig(L); lams=diag(D);
idx=find(abs(lams-lams(2))<1e-8);
v1=P(:,idx(1)); v2=P(:,idx(2));
fprintf('v1='); disp(v1'); fprintf('v2='); disp(v2');
```

**(b)** Try $u(\theta) = \cos\theta\, v_1 + \sin\theta\, v_2$ for several values of $\theta$; each produces a different sign-partition. At least two distinct partitions will emerge (corresponding to the two "natural" 3-vs-3 splits).

**(c)** The graph has the symmetry $\sigma : i \mapsto i + 3 \pmod 6$ (swapping the two triangles). This symmetry sends each eigenvector $v$ to $-v$ (the anti-symmetric mode). Therefore for any Fiedler vector $u = \cos\theta\, v_1 + \sin\theta\, v_2$:

$$u_{i+3} = -u_i \quad \text{for } i = 1, 2, 3.$$

Since $u_i$ and $u_{i+3}$ have opposite signs (and neither is 0 generically), vertices $i$ and $i + 3$ always fall in opposite subgraphs. □ □

**Exercise 12.13.** $K_n$: what are the eigenvalues of the Laplacian?

*Solution.* $L_{K_n} = nI - J$ where $J = \mathbf{1}\mathbf{1}^T$. Eigenvectors of $J$: $\mathbf{1}$ (eigenvalue $n$) and any vector orthogonal to $\mathbf{1}$ (eigenvalue 0). So eigenvalues of $nI - J$: $n - n = 0$ (once) and $n - 0 = n$ ($n - 1$ times).

**Guess:** $\lambda(L_{K_n}) = \{0^{(1)}, n^{(n-1)}\}$.

Verification:

```
for n=[3 4 5 6], A=ones(n,n)-eye(n); L=diag(sum(A))-A;
fprintf('K_%d: ',n); fprintf('%.1f ',sort(eig(L))'); fprintf('\n'); end
```

□

**Exercise 12.14.** 11-vertex graph (from notes). (a) Find Fiedler vector. (b) Group sorted values into 3 subsets. (c) Partition into 3 subgraphs. (d) Draw. (e) Structural insights.

*Solution.* **(a)**

```
A=make_adj(edges,11); L=make_lap(A);
[fv,lam2]=fiedler_vec(L);
[fv_s,idx]=sort(fv);
fprintf('Sorted Fiedler entries:\n');
for i=1:11, fprintf('  v%d: %.4f\n',idx(i),fv_s(i)); end
```

**(b)** The sorted Fiedler values split into 3 groups separated by gaps:

- **Group A** (most negative): vertices with $v_i \ll 0$
- **Group B** (near zero): vertices with $v_i \approx 0$ (bridge/central)
- **Group C** (most positive): vertices with $v_i \gg 0$

Identify the two largest gaps in the sorted sequence to determine group boundaries.

**(c)** Assign each vertex to the group from (b). The three-way partition minimizes the number of cross-edges.

**(d)** Draw each group as a subgraph; draw dashed edges for cross-group connections.

**(e)** The three groups reveal which subsets of the 11-vertex graph are most internally cohesive, and the dashed cross-edges show the bottleneck connections. □

**Exercise 12.15.** Path graph $1 - 2 - \cdots - n$. (a) Predict Fiedler Method outcome before calculating; justify. (b) Check hypothesis for several values of $n$.

*Solution.* **(a) Prediction.**

The path $P_n$ is a chain: vertices $1, 2, \ldots, n$ connected as $i$–$(i+1)$. There is no cycle; the graph is "one-dimensional." The natural partition into two connected halves is $\{1, \ldots, \lfloor n/2 \rfloor\}$ and $\{\lceil n/2 \rceil + 1, \ldots, n\}$, cutting exactly one edge.

**Hypothesis:** The Fiedler vector of $P_n$ has entries monotonically increasing (or decreasing) from vertex 1 to $n$, with the cut at the midpoint (one cut-edge).

*Justification:* The Laplacian of $P_n$ is the tridiagonal matrix $L_{ij} = -1$ for $|i - j| = 1$, $L_{ii} = 2$ for $1 < i < n$, and $L_{11} = L_{nn} = 1$. Its eigenvectors are known analytically: the $k$-th eigenvector has entries $v_i^{(k)} = \cos\left(\frac{(i-\frac{1}{2})(k-1)\pi}{n}\right)$. The Fiedler vector ($k = 2$) is $v_i \propto \cos\left(\frac{(i-\frac{1}{2})\pi}{n}\right)$, which is strictly monotone in $i$. The sign change occurs near $i = n/2$.

**(b) Verification.**

```
for n=[4 5 6 7 8]
  edges=[(1:n-1)',(2:n)'];
  A=make_adj(edges,n); L=make_lap(A);
  [fv,lam2]=fiedler_vec(L);
  fprintf('n=%d, lam2=%.4f, Fiedler=',n,lam2);
  fprintf('%.3f ',fv'); fprintf('\n');
end
```

For $n = 4$: $\lambda_2 = 2 - \sqrt{2} \approx 0.586$, $v \approx (-0.65, -0.27, 0.27, 0.65)$. Sign change between positions 2 and 3 $\Rightarrow$ partition $\{1, 2\} \mid \{3, 4\}$, cut = edge 2–3. ✓

For odd $n$: the middle vertex has $v_{(n+1)/2} \approx 0$ and can go in either group; the cut is still 1 edge. □

**Exercise 12.16.** 14-vertex graph (from notes). (a) Find Fiedler vector. (b) Separate into 3 groups. (c) Partition. (d) Draw. (e) Which vertex is most critical? (f) Structural insights.

*Solution.*

```
A=make_adj(edges,14); L=make_lap(A);
[fv,lam2]=fiedler_vec(L);
[fv_s,ord]=sort(fv);
fprintf('Sorted: '); fprintf('v%d:%.3f ',ord,fv_s); fprintf('\n');
```

**(b)** Inspect the sorted values for two large gaps. Label the three resulting groups A (negative), B (near-zero bridge), C (positive).

**(c)** The partition is $G_A \mid G_B \mid G_C$. Edges between groups are the cut-edges.

**(d)** Draw each subgraph separately; dashed lines show inter-group edges.

**(e)** The *most critical* vertex is the one in group B (bridge) with the Fiedler entry closest to 0 *and* the highest degree connecting to both other groups. Removing it would maximally disconnect the graph. If group B has only one vertex, that vertex is the critical articulation point.

**(f)** The three-group partition reveals hub-and-spoke structure or community structure. Look at whether the bridge group (B) is a single articulation point or a small cluster. □

**Exercise 12.17.** 13-vertex social network. (a) Use Fiedler Method to identify group most strongly connected to Person 10. (b) Why is the method ineffective?

*Solution.*

```
A=make_adj(edges,13); L=make_lap(A);
[fv,lam2]=fiedler_vec(L);
sign10=sign(fv(10));
group_with_10=find(sign(fv)==sign10);
fprintf('Group containing P10: '); disp(group_with_10');
```

**(a)** The Fiedler partition assigns Person 10 to the group with the same sign Fiedler entry. This group is the one identified as "most strongly connected" to P10 globally.

**(b) Why ineffective:** The Fiedler method is a *global* spectral method – it minimizes the total cut and finds a balanced partition of the entire graph. It does not find the local neighborhood of a specific vertex. Person 10 may have close friends (short graph-distance neighbors) who end up in the *opposite* Fiedler partition if they happen to be on the "wrong" side of the global minimum-cut. A better approach for finding P10's strongest connections is BFS/DFS, clustering coefficients, or ego-network analysis, not the Fiedler partition. □ □

## SP3 Quiz: Graph Theory Problems

**Exercise SP3 Quiz 1.** Give examples of graphs $\Gamma_1$ and $\Gamma_2$ with the *same number of triangles* but which are *not isomorphic*. Fully justify using a theorem or definition.

*Solution.* **Example.**

- $\Gamma_1$: Two disjoint triangles, $K_3 \sqcup K_3$ (6 vertices, 6 edges, **2 triangles**, not connected).
- $\Gamma_2$: Triangular prism graph (6 vertices, 9 edges, **2 triangles**, connected, 3-regular). Its edges are $\{1\text{-}2, 2\text{-}3, 1\text{-}3, 4\text{-}5, 5\text{-}6, 4\text{-}6, 1\text{-}4, 2\text{-}5, 3\text{-}6\}$.

**Same triangle count:** $\Gamma_1$ has triangles $\{1, 2, 3\}$ and $\{4, 5, 6\}$ (2 total). $\Gamma_2$ has triangles $\{1, 2, 3\}$ and $\{4, 5, 6\}$ (2 total). ✓

**Not isomorphic:** By Definition 12.2.0.3, $\Gamma_1$ is *not connected* (it has two separate pieces). $\Gamma_2$ *is connected*. Connectivity is preserved under isomorphism (an isomorphism is a bijection on vertices preserving adjacency, so paths are preserved). Therefore $\Gamma_1 \ncong \Gamma_2$. □ □

**Exercise SP3 Quiz 2.** Give examples of: (a) a graph that is not simple but has a simple subgraph; (b) a graph that is not connected but has a connected subgraph.

*Solution.* **(a) Non-simple graph with a simple subgraph.**

*Example:* Let $G$ have vertices $\{1, 2\}$ and two parallel edges between them (a multigraph, violating "no multiple edges" in Definition 12.2.0.2, hence not simple). The subgraph retaining only *one* of the two edges is $K_2$ (the complete graph on 2 vertices), which is simple.

*Note:* Any non-simple graph obtained by adding loops or multi-edges to a simple graph trivially contains the original simple graph as a subgraph.

**(b) Non-connected graph with a connected subgraph.**

*Example:* $G = K_3 \sqcup K_2$ (a triangle on vertices $\{1, 2, 3\}$ and a separate edge on vertices $\{4, 5\}$). $G$ is not connected (vertices 1 and 4 are in different components). The subgraph $K_3$ (induced on $\{1, 2, 3\}$) is connected. □ □

## SP3 Exam: Additional Graph Theory Problems

**Exercise SP3 Exam: Graph Isomorphism.** (a) How many $3 \times 3$ permutation matrices are there? (b) Prove $K_3 \ncong P_3$ by checking all permutation matrices. (c) Use the triangle theorem.

*Solution.* **(a)** $3! = 6$.

**(b)** $A_1 = K_3$ (triangle), $A_2 = P_3$ (path). None of the 6 permutations gives $PA_2P^T = A_1$.

```
A1=[0 1 1;1 0 1;1 1 0]; A2=[0 1 0;1 0 1;0 1 0];
found=false;
for k=1:6, P=eye(3)(perms([1 2 3])(k,:),:);
  if all(all(P*A2*P'==A1)), found=true; disp('equal'); end; end
if ~found, disp('Not isomorphic: no permutation works.'); end
```

Output: `Not isomorphic: no permutation works.`

**(c)** By Theorem 12.3.0.2 (triangles $= \frac{1}{6}\operatorname{tr}(A^3)$): $K_3$ has $\operatorname{tr}(A_1^3)/6 = 6/6 = 1$ triangle; $P_3$ has $\operatorname{tr}(A_2^3)/6 = 0/6 = 0$ triangles. Since isomorphic graphs have the same number of triangles, $K_3 \ncong P_3$. □ □

**Exercise SP3 Exam: Planar and Regular Subgraphs.** (a) Is it possible to find a *planar* graph with a *nonplanar* subgraph?
(b) Is it possible to find a *regular* graph with a *nonregular* subgraph?
Give an example or explain why none exist.

*Solution.* **(a) No, a planar graph cannot have a nonplanar subgraph.**

*Proof:* By Kuratowski's theorem, a graph is *nonplanar* if and only if it contains a subdivision of $K_5$ or $K_{3,3}$ as a subgraph. If $H$ is a subgraph of $G$ and $H$ is nonplanar, then $H$ contains a subdivision of $K_5$ or $K_{3,3}$, which is also a subgraph of $G$. But then $G$ is nonplanar – contradiction. □

*Intuition:* Planarity is a "downward closed" property: removing vertices or edges from a planar graph always leaves a planar graph.

**(b) Yes, a regular graph can have a nonregular subgraph.**

*Example:* $G = K_4$ (complete graph on 4 vertices) is 3-regular. Consider the subgraph $H$ obtained by removing the single edge $\{1, 2\}$:

$$\deg_H(1) = 2, \quad \deg_H(2) = 2, \quad \deg_H(3) = 3, \quad \deg_H(4) = 3.$$

$H$ has vertices of degree 2 and degree 3, so $H$ is not regular.

$H$ is a valid subgraph of $K_4$ (retaining all 4 vertices, removing one edge). □

*Note:* Regularity is *not* a hereditary (subgraph-monotone) property, unlike planarity. □

**True/False (from Sol. 12):**

**"Same adjacency matrix $\Rightarrow$ isomorphic"**: **True**. Identical adjacency matrices (with the same vertex labeling) means the graphs have exactly the same edge structure, hence $P = I$ witnesses isomorphism. $\square$

**"Same degree matrix $\Rightarrow$ isomorphic"**: **False**. Counterexample: $C_6$ (6-cycle, every vertex degree 2) and $2K_3$ (two disjoint triangles, every vertex degree 2) both have degree sequence $(2, 2, 2, 2, 2, 2)$ but are not isomorphic ($C_6$ is connected, $2K_3$ is not).

# Chapter 6

# Markov Chains — Solutions

**Exercise 6.1.** Population diagram (3 states): (a) transition matrix; (b) asymmetry; (c) 1 and 2 iterations; (d) steady state; (e) smallest $k$ for 4-decimal accuracy.

*Solution.* $T = \begin{pmatrix} 0.80 & 0.05 & 0.25 \\ 0.12 & 0.90 & 0.05 \\ 0.08 & 0.05 & 0.70 \end{pmatrix}$, $\mathbf{x}_0 = (0.6, 0.2, 0.2)^T$.

**(b)** $T$ not symmetric $\Rightarrow$ movement rates differ by direction.

**(c)** $\mathbf{x}_1 = T\mathbf{x}_0 = (0.540, 0.262, 0.198)^T$; $\mathbf{x}_2 = T^2\mathbf{x}_0 \approx (0.497, 0.303, 0.200)^T$.

**(d)** Solve $(T - I)\mathbf{x}^* = \mathbf{0}$ with $\sum x_i = 1$: $\mathbf{x}^* \approx (0.416, 0.416, 0.168)^T$.

**(e)**

```
T=[0.80 0.05 0.25;0.12 0.90 0.05;0.08 0.05 0.70];
x=T^1*[0.6;0.2;0.2]; xs=steady_state(T); k=1;
while max(abs(x-xs))>5e-5, k=k+1; x=T*x; end; fprintf('k=%d\n',k);
```

□

**Exercise 6.2.** $6 \times 6$ transition matrix: (a) diagram; (b) regular?

*Solution.* $T = \begin{pmatrix} 0.9 & 0 & 0.05 & 0 & 0.15 & 0.05 \\ 0 & 0.8 & 0.05 & 0.2 & 0.05 & 0.05 \\ 0.01 & 0.1 & 0.5 & 0 & 0 & 0.05 \\ 0.02 & 0.1 & 0.1 & 0.75 & 0 & 0.05 \\ 0.07 & 0 & 0.1 & 0 & 0.7 & 0.05 \\ 0 & 0 & 0.2 & 0.05 & 0.1 & 0.75 \end{pmatrix}$.

**(b)** Regular, since $T^2$ has all positive entries. All states communicate (every state can reach every other state in at most 2 steps). □

**Exercise 6.3.** Construct a 3-state regular symmetric transition matrix.

*Solution.* $T = \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{pmatrix}$.

Symmetric (equal migration rates), regular (all entries positive). Steady state: $\mathbf{x}^* = (1/3, 1/3, 1/3)^T$ by symmetry. □

**Exercise 6.4.** Large diagram: transition matrix, entries of $T^k$, zero entries for all $k$.

*Solution.* $T$ is block upper-triangular (states 4,5,6 cannot reach states 1,2,3).

**(a)** Entry $(3, 2)$ of $T^1$: read directly from diagram. **(b)** Entry $(3, 2)$ of $T^2$: $\sum_\ell T_{3\ell} T_{\ell 2}$. **(c)** Entry $(1, 3)$ of $T^2$: 0 if 3 cannot reach 1 in 2 steps. **(d)** Smallest $k$ for nonzero $(5, 3)$: path length

$3 \to 4 \to 5$ or $3 \to 5$. **(e)** $(i, j)$ zero for all $k$: if state $j$ is in the "absorbing" block unreachable from $i$ *or* vice versa. **(f)** Long term: population drains into states 4,5,6. □

**Exercise 6.5.** Transition matrix with $T^7 = I$ but $T^k \neq I$ for $k < 7$.

*Solution.* Use a 7-state chain cycling $1 \to 2 \to 3 \to \cdots \to 7 \to 1$ deterministically: $T_{i+1,i} = 1$ (and $T_{1,7} = 1$), 0 elsewhere. Then $T^7 = I$. This is not regular (period 7). □

**Exercise 6.6.** $5 \times 5$ non-regular $T$ with all states converging to a unique steady state.

*Solution.* States $\{1, 2, 3\}$ form a regular sub-chain; states 4,5 always transition into $\{1, 2, 3\}$. Example:

$$T = \begin{pmatrix} 0.7 & 0.1 & 0.2 & 0.5 & 0.5 \\ 0.1 & 0.8 & 0.1 & 0.3 & 0.3 \\ 0.2 & 0.1 & 0.7 & 0.2 & 0.2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Non-regular (states 4,5 are absorbing-like and $T_{i4} = T_{i5} = 0$ for $i > 3$). But all states eventually drain to $\{1, 2, 3\}$, which has a unique steady state. □

**Exercise 6.7.** 3-state diagram: T, iterations, steady state.

*Solution.* $T = \begin{pmatrix} 0.70 & 0.05 & 0.02 \\ 0.28 & 0.85 & 0.03 \\ 0.02 & 0.10 & 0.95 \end{pmatrix}$, $\mathbf{x}_0 = (0.1, 0, 0.9)^T$.

**(a) Transition matrix:** as above.

**(b)** $\mathbf{x}_1 = T\mathbf{x}_0 = (0.088, 0.113, 0.799)^T$; compute further iterations programmatically.

**(c)** Steady state: solve $(T - I)\mathbf{x}^* = \mathbf{0}$, $\sum x_i = 1$.

```
T=[0.70 0.05 0.02;0.28 0.85 0.03;0.02 0.10 0.95];
x0=[0.1;0;0.9]; [V,D]=eig(T); idx=find(abs(diag(D)-1)<1e-9);
xs=real(V(:,idx)); xs=abs(xs)/sum(abs(xs));
```

□

**Exercise 6.8.** Find eigenvalues of $T = \begin{pmatrix} a & 0 & 1-c \\ 1-a & b & 0 \\ 0 & 1-b & c \end{pmatrix}$ (columns sum to 1).

*Solution.* $\lambda = 1$ is always an eigenvalue (columns sum to 1). The characteristic polynomial gives:

$$\mathrm{char}(T) = (\lambda - 1)\big(\lambda^2 - (a + b + c - 1)\lambda + (ab + bc + ca - a - b - c + 1)\big).$$

The other two eigenvalues are roots of the quadratic factor. Their values depend on $a, b, c$. □

**Exercise 6.9.** Find a $2 \times 2$ regular $T$ and $\mathbf{x}_0$ where $\mathbf{x}_{1000}$ still differs from $\mathbf{x}^*$ at the first decimal place.

*Solution.* Slow convergence requires $|\lambda_2| = |a+b-1|$ close to 1. Take $a = 0.9999$ and $b = 0.9999$:

$$T = \begin{pmatrix} 0.9999 & 0.0001 \\ 0.0001 & 0.9999 \end{pmatrix}.$$

$\lambda_2 = 0.9998$. $\lambda_2^{1000} = 0.9998^{1000} \approx 0.819$. Deviation is $\approx 0.5 \times 0.819 \approx 0.41$, so first decimal place still differs. $\mathbf{x}_0 = (1, 0)^T$, $\mathbf{x}^* = (0.5, 0.5)^T$. □

**Exercise 6.10.** Transition matrix for a disconnected diagram.

*Solution.* Two disconnected 3-state chains:

$$T = \begin{pmatrix} 0.7 & 0.1 & 0.2 & 0 & 0 & 0 \\ 0.1 & 0.8 & 0.1 & 0 & 0 & 0 \\ 0.2 & 0.1 & 0.7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.6 & 0.3 & 0.1 \\ 0 & 0 & 0 & 0.2 & 0.5 & 0.3 \\ 0 & 0 & 0 & 0.2 & 0.2 & 0.6 \end{pmatrix}.$$

□

**Exercise 6.11.** Disconnected $T$: one component with steady state, one without.

*Solution.* Component 1: regular (has steady state). Component 2: $T_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (non-regular, oscillates). Block-diagonal:

$$T = \begin{pmatrix} T_1 & 0 \\ 0 & T_2 \end{pmatrix}.$$

□

**Exercise 6.12.** Define: probability vector, transition matrix, regular transition matrix.

*Solution.* **Probability vector:** A non-negative vector whose entries sum to 1.

**Transition (stochastic) matrix:** A matrix whose columns are all probability vectors. Entry $T_{ij}$ = fraction of population moving from state $j$ to state $i$ per step.

**Regular transition matrix:** A transition matrix $T$ such that $T^k$ has all strictly positive entries for some integer $k \geq 1$. □

**Exercise 6.13.** For any $n$: can some entry of $T^k$ be 0 for $k < n$ but nonzero at $k = n$?

*Solution.* **Yes.** Use the $n$-cycle transition matrix: $T_{i+1,i} = 1$ cyclically. Then $(T^k)_{1,1} = 0$ for $1 \leq k < n$ (no closed walk of length $< n$ starting and ending at state 1), but $(T^n)_{1,1} = 1$. □

**Exercise 6.14.** Two-state: (a) transition matrix; (b) steady state; (c) restrictions.

*Solution.* $T = \begin{pmatrix} \alpha & \beta \\ 1-\alpha & 1-\beta \end{pmatrix}$ where $\alpha$ = stay at 1, $1-\beta$ = stay at 2.

**(b)** Steady state: $(T - I)\mathbf{x} = \mathbf{0}$, $x_1 + x_2 = 1$:

$$x_1^* = \frac{\beta}{(1-\alpha)+\beta}, \quad x_2^* = \frac{1-\alpha}{(1-\alpha)+\beta}.$$

**(c)** If $\alpha = 1, \beta = 1$ (no movement): $T = I$, any probability vector is a steady state ($\infty$ many). If $\alpha = 0, \beta = 0$: $T = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$, state 2 is absorbing – unique steady state $\mathbf{x}^* = (0,1)^T$ but not "limiting" from all starts. Without the restrictions, steady states can be non-unique or non-convergent. □

**Exercise 6.15.** If $T^k$ has all nonzero entries, show $T^j$ does too for $j > k$.

*Solution.* Since $T$ is a transition matrix, all entries $T_{ij} \geq 0$ and each column sums to 1 (so at least one entry per column is $> 0$). For $j = k+1$: $(T^{k+1})_{ij} = \sum_\ell T_{i\ell}(T^k)_{\ell j}$. Since $(T^k)_{\ell j} > 0$ for all $\ell, j$ and $T_{i\ell} \geq 0$ with $\sum_\ell T_{i\ell} = 1 > 0$, at least one term is positive. So $(T^{k+1})_{ij} > 0$. By induction, $T^j$ has all positive entries for all $j \geq k$. □ □

# Chapter 7

# Google PageRank — Solutions

**Exercise 7.1.** Build the PageRank transition matrix for a small internet and find the ranking vector.

*Solution.* Example: P1→P2, P1→P3, P2→P3, P3→P1. Link matrix:

$$S = \begin{pmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 0 \\ 1/2 & 1 & 0 \end{pmatrix}.$$

Google matrix with $d = 0.85$, $n = 3$:

$$T = \frac{0.15}{3} J + 0.85 S = \begin{pmatrix} 0.05 & 0.05 & 0.9 \\ 0.475 & 0.05 & 0.05 \\ 0.475 & 0.9 & 0.05 \end{pmatrix}.$$

```
links=[1 2;1 3;2 3;3 1]; T=make_gpmatrix(links,3); pr=pagerank_vec(T);
```

**Output:** $\mathbf{x}^* \approx (0.278, 0.155, 0.567)^T$. Ranking: P3 > P1 > P2. □

**Exercise 7.2.** Construct the PageRank matrix from a web diagram with 4 pages; rank them.

*Solution.*

```
links=[1 2;2 1;2 3;3 4;4 1;4 2]; T=make_gpmatrix(links,4); pr=pagerank_vec(T);
```

Rankings depend on the diagram. The page receiving the most links from high-ranked pages wins. □

**Exercise 7.3.** How does adding a dangling node affect the matrix?

*Solution.* A dangling node $j$ (no outbound links) contributes column $j = (1/n, \ldots, 1/n)^T$ to $S$ – a uniform jump. The PageRank matrix is unchanged in structure; the dangling node merely distributes its "page traffic" equally. Adding dangling nodes slightly dilutes the link-structure information but the matrix remains regular. □

**Exercise 7.4.** $P1 \leftrightarrow P2$: both have equal PageRank? Prove by symmetry.

*Solution.* $T = \frac{0.15}{2} J + 0.85 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0.075 & 0.925 \\ 0.925 & 0.075 \end{pmatrix}$.

Steady state: $T\mathbf{x}^* = \mathbf{x}^*$. Symmetry gives $x_1^* = x_2^* = 0.5$. □

**Exercise 7.5.** Can adding outbound links change your PageRank?

*Solution.* **No.** Outbound links determine *where your traffic goes*, not how much you receive. Adding an outbound link to page $j$ changes column $j$ of $S$ (and hence $T$), but this changes the steady state *of other pages*, not necessarily yours.

In practice, adding outbound links slightly reduces your traffic concentration (your column sends traffic elsewhere), which can reduce your PageRank marginally. □

**Exercise 7.6.** Chain internet $P1 \to P2 \to P3$ (P3 dangling): build $T$, rank.

*Solution.* $n = 3$. P3 is dangling, so $S_{j3} = 1/3$ for all $j$:

$$S = \begin{pmatrix} 0 & 0 & 1/3 \\ 1 & 0 & 1/3 \\ 0 & 1 & 1/3 \end{pmatrix}, \quad T = \frac{0.15}{3} J + 0.85S.$$

```
links=[1 2;2 3]; T=make_gpmatrix(links,3); pr=pagerank_vec(T);
```

Ranking: P3>P2>P1 (P3 receives a link from P2 which receives from P1; traffic accumulates further down the chain, plus P3 redistributes uniformly). □

**Exercise 7.7.** Effect of damping factor $d$ on rankings.

*Solution.* As $d \to 1$: $T \to S$ (pure link structure). Isolated pages with no inbound links die; all traffic follows links exactly. Unique steady state may not exist if $S$ is not regular.

As $d \to 0$: $T \to \frac{1}{n}J$. $T\mathbf{x} = \frac{1}{n}J\mathbf{x} = \frac{1}{n}\mathbf{1} \to \mathbf{u} = (1/n, \ldots, 1/n)^T$. Uniform – link structure has no effect.

$d = 0.85$ is empirically chosen to weight link structure heavily while guaranteeing convergence. □

**Exercise 7.8.** Fully connected $n$-page internet: show $\mathbf{u} = (1/n, \ldots, 1/n)^T$ is the steady state.

*Solution.* Each page links to all $n-1$ others, so $S_{ij} = 1/(n-1)$ for $i \neq j$ and $S_{ii} = 0$. The all-ones matrix $J$ gives $J\mathbf{u} = \mathbf{1}$. Then:

$$T\mathbf{u} = \frac{0.15}{n} J\mathbf{u} + 0.85S\mathbf{u} = \frac{0.15}{n} \cdot n\mathbf{u} + 0.85 \cdot \frac{n}{n-1}\mathbf{1} \cdot \frac{1}{n}.$$

Wait: $S\mathbf{u} = \frac{1}{n-1}(J - I)\mathbf{u} = \frac{1}{n-1}(\mathbf{1} - \mathbf{u}) = \frac{1}{n-1} \cdot \frac{n-1}{n}\mathbf{1} = \frac{1}{n}\mathbf{1} = \mathbf{u}$. So $T\mathbf{u} = 0.15\mathbf{u} + 0.85\mathbf{u} = \mathbf{u}$. ✓
□ □

**Exercise 7.9.** Which of the following is more effective at raising your PageRank: (a) getting a link from a high-ranked page, or (b) getting many links from low-ranked pages?

*Solution.* **(a) One high-ranked link is generally better.** A high-ranked page passes a large share of its traffic. A low-ranked page passes very little. Even many low-ranked inbound links may not equal one from a page that itself has high PageRank.

However, there's a trade-off: if a high-ranked page links to many pages, each gets only a small share ($1/k$ of the high rank). So the answer also depends on how many other outbound links the high-ranked page has. □

**Exercise 7.10.** Disconnected internet without random jump: why fails? Fix?

*Solution.* Without the random-jump term: $T = S$. If the internet is disconnected, $S$ is block upper-triangular (or has structural zeros). $S$ may not be regular $\Rightarrow$ no unique steady state $\Rightarrow$ PageRank is not uniquely defined.

**Fix:** The random-jump term $\frac{1-d}{n}J$ adds $(1-d)/n > 0$ to every entry, making $T$ strictly positive. Hence $T^1$ has all positive entries $\Rightarrow$ $T$ is regular $\Rightarrow$ unique steady state guaranteed. □

**Exercise 7.11.** Build and rank a 6-page internet with the given link structure.

*Solution.*

```
links=[1 2;1 4;2 3;3 1;4 5;5 6;6 4;2 5];
T=make_gpmatrix(links,6,0.85); pr=pagerank_vec(T);
fprintf('PageRank: '); fprintf('%.4f ',pr); fprintf('\n');
[~,order]=sort(pr,'descend');
fprintf('Ranking: '); fprintf('P%d ',order); fprintf('\n');
```

**Output:** Ranking varies by link structure (output shown in compiled code). □

# Chapter 8

# Singular Value Decomposition — Solutions

**Exercise 8.1.** Find the SVD of each matrix. First compute $AA^T$ and $A^TA$, find eigenpairs, then verify with technology.

*Solution.* **Method:** (1) Compute $AA^T$ and $A^TA$; (2) find eigenvalues $\sigma_i^2$ (shared, nonzero); (3) for each $\sigma_i^2$, get unit eigenvector $\bar{u}_i$ of $AA^T$ and set $\bar{v}_i = \frac{1}{\sigma_i}A^T\bar{u}_i$; (4) build $U$, $\Sigma$, $V$.

**(a)** $A = \begin{pmatrix} 1 & 2 & -3 \\ 0 & 1 & 1 \\ 1 & 2 & 5 \\ -1 & 0 & 2 \end{pmatrix}$ $(4 \times 3)$.

$AA^T$ is $4 \times 4$, $A^TA$ is $3 \times 3$. Eigenvalues of $A^TA$: $\{18, 6, 0\}$ (or compute both; they share nonzero eigenvalues). So $\sigma_1 = \sqrt{18}$, $\sigma_2 = \sqrt{6}$, $\sigma_3 = 0$ (only 2 nonzero).

$\Sigma = \mathrm{diag}(\sqrt{18}, \sqrt{6}, 0)$ in the $4 \times 3$ rectangular form (last row zeros).

```
A = [1 2 -3; 0 1 1; 1 2 5; -1 0 2];
[U,S,V] = svd(A);
fprintf('Singular values: '); disp(diag(S)');
fprintf('U:\n'); disp(U);
fprintf('V:\n'); disp(V);
```

Output: $\sigma_1 \approx 4.2426$, $\sigma_2 \approx 2.4495$, $\sigma_3 \approx 0$. Check: $A \approx U\Sigma V^T$.

**(b)** $A = \begin{pmatrix} -1 & 0 & 2 & 2 & 2 \\ 0 & 2 & 3 & 0 & 1 \\ 1 & 2 & -2 & 1 & 2 \end{pmatrix}$ $(3 \times 5)$.

$A^TA$ is $5 \times 5$; eigenvalues include 0 (multiplicity 2). Nonzero: $\sigma_1^2$, $\sigma_2^2$, $\sigma_3^2$. Compute via technology.

```
A = [-1 0 2 2 2; 0 2 3 0 1; 1 2 -2 1 2];
[U,S,V] = svd(A);
fprintf('Singular values: '); disp(diag(S)');
```

**(c)** Same as (a) — matrix is identical.

**(d)** $A = \begin{pmatrix} 0.1 & 0.2 & 0.9 & 0.3 \\ 0.9 & 0.2 & 0 & 0.2 \\ 0.2 & 0.2 & 0.3 & 0.1 \\ 0 & 0.3 & 0.7 & 0.6 \end{pmatrix}$ $(4 \times 4)$.

```
A = [0.1 0.2 0.9 0.3; 0.9 0.2 0 0.2; 0.2 0.2 0.3 0.1; 0 0.3 0.7 0.6];
[U,S,V] = svd(A);
fprintf('Singular values: '); disp(diag(S)');
```

All four singular values are positive (matrix has full rank). Verify $A = U\Sigma V^T$ numerically. □

# Chapter 9

# Image Compression — Solutions

**Exercise 10.1.** Consider the $4 \times 4$ image (shades 0, 0.5, 1). (a) Enter into matrix $A$. (b) Find SVD. (c) Simplify with 3, 2, 1 singular values; draw best representation.

*Solution.* **(a)** From the image in the text (black/white/grey pattern), a plausible $4 \times 4$ matrix using only 0, 0.5, 1:

$$A = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

(Adjust entries to match the actual figure if different.)

**(b)**

```
A = [1 0 1 1; 1 1 0 1; 0 1 0 1; 0 1 1 0];
[U,S,V] = svd(A);
svals = diag(S);
fprintf('Singular values: '); disp(svals');
```

Four singular values (e.g., $\sigma_1 \approx 2.4$, $\sigma_2 \approx 1.2$, $\sigma_3 \approx 0.8$, $\sigma_4 \approx 0.4$).

**(c)** For $k = 3, 2, 1$: set $S(i, i) = 0$ for $i > k$, then $A' = U * S * V'$. Clamp values to $[0, 1]$ for display (or leave as-is per notes). Draw the resulting $4 \times 4$ grid: darker = lower value, lighter = higher.

```
for k = [3 2 1]
  Sp = S; Sp((k+1):4, (k+1):4) = 0;
  Ap = U*Sp*V';
  fprintf('k=%d, variance preserved: %.4f\n', k, ...
    sum(svals(1:k).^2)/sum(svals.^2));
  disp(Ap);
end
```

□

**Exercise 10.2.** Consider the $5 \times 5$ image (shades 0, 0.5, 1). (a) Enter into matrix $A$. (b) Find SVD. (c) Simplify with 4, 3, 2, 1 singular values; draw.

*Solution.* **(a)** From the image in the text:

$$A = \begin{pmatrix} 1 & 0.5 & 0 & 1 & 1 \\ 0.5 & 1 & 0.5 & 0 & 1 \\ 0 & 0.5 & 1 & 0.5 & 0 \\ 1 & 0 & 0.5 & 1 & 0.5 \\ 1 & 1 & 0 & 0.5 & 1 \end{pmatrix}$$

(Adjust to match the actual figure.)

**(b)–(c)** Same procedure as 10.1: `svd(A)`, then zero out smallest singular values for $k = 4, 3, 2, 1$, reconstruct, and draw.

```
A = [1 0.5 0 1 1; 0.5 1 0.5 0 1; 0 0.5 1 0.5 0; 1 0 0.5 1 0.5; 1 1 0 0.5 1];
[U,S,V] = svd(A);
svals = diag(S);
for k = [4 3 2 1]
  Sp = S; for i=k+1:5, Sp(i,i)=0; end
  Ap = U*Sp*V';
  fprintf('k=%d: var=%.4f\n', k, sum(svals(1:k).^2)/sum(svals.^2));
end
```

□

**Exercise 10.3.** Load a square image ($\sim 200 \times 200$). (a) Load, grayscale, scale to $[0, 1]$. (b) SVD. (c) Count singular values. (d)–(f) Preserve 75%, 50%, 25% of singular values; report quality %. (g) Min $k$ for 99.9% quality; compression ratio; print result.

*Solution.*

```
A = imread('image.jpg');
A = rgb2gray(A); A = im2double(A);
A = A - min(A(:)); A = A / max(A(:));

[U,S,V] = svd(A);
svals = diag(S);
n = length(svals);
fprintf('(c) Singular values: %d\n', n);

for pct = [0.75 0.50 0.25]
  k = round(pct * n);
  var_pres = sum(svals(1:k).^2) / sum(svals.^2);
  fprintf('Preserve %.0f%% of values: k=%d, quality=%.4f%%\n', ...
    pct*100, k, var_pres*100);
end

target = 0.999;
cum = cumsum(svals.^2);
tot = cum(end);
k_min = find(cum/tot >= target, 1);
var_at_k = cum(k_min)/tot;
comp_ratio = 2*k_min*n / n^2;
fprintf('(g) Min k for 99.9%%: %d, quality=%.4f%%, compression=%.2f\n', ...
  k_min, var_at_k*100, comp_ratio);

Sp = S; Sp((k_min+1):end, (k_min+1):end) = 0;
Ap = U*Sp*V';
imshow(Ap, 'border', 'tight');
```

□

**Exercise 10.5.** For each $10 \times 10$ image with given singular values, find (a)–(d) the minimum $k$ for 99.9% variance and the compression ratio.

*Solution.* Quality $= \frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_{10}^2}$. Find smallest $k$ with $\geq 0.999$. Compression ratio $= 2k/n = 2k/10 = k/5$.

**(a)** $\sigma = [80.26, 63.35, 20.59, 8.47, 2.88, 1.68, 0.80, 0.69, 0.66, 0.65]$.

```
s = [80.2608 63.3520 20.5871 8.4696 2.8841 1.6763 0.7962 0.6926 0.6553 0.6520];
cum = cumsum(s.^2); k = find(cum/cum(end) >= 0.999, 1);
fprintf('k=%d, ratio=%.2f\n', k, 2*k/10);  % ratio = 2k/n
```

$k = 2$ gives $\approx 0.9997$; ratio $= 0.4$.

**(b)** $\sigma = [138.55, 49.52, 16.59, 4.94, 3.24, 1.52, 1.20, 1.03, 1.02, 0.98]$. $k = 2$ suffices; ratio $= 0.4$.

**(c)** $\sigma = [446.16, 163.42, 43.89, 13.80, 4.74, 1.34, 1.05, 0.64, 0.45, 0.45]$. $k = 2$; ratio $= 0.4$.

**(d)** $\sigma = [209.49, 28.19, 22.87, 11.83, 6.33, 2.78, 1.20, 0.93, 0.88, 0.87]$. $k = 3$ or 4; ratio $= 0.6$ or 0.8. □

**Exercise 10.6.** For an $m \times n$ image simplified with $k$ singular values: (a) compression ratio; (b) criteria for worth doing.

*Solution.* **(a) Storage.**

We store:

- $k$ left singular vectors $\bar{u}_1, \ldots, \bar{u}_k$, each in $\mathbb{R}^m$: $km$ values.
- $k$ right singular vectors $\bar{v}_1, \ldots, \bar{v}_k$, each in $\mathbb{R}^n$: $kn$ values.
- $k$ singular values: $k$ values (often negligible for large $m, n$).

Total: $km + kn = k(m + n)$ values (ignoring the $k$ singular values for the ratio).

Original image: $mn$ values.

**Compression ratio:**

$$\boxed{\frac{k(m+n)}{mn} = k\left(\frac{1}{n} + \frac{1}{m}\right).}$$

**(b) Worth doing when** $k(m + n) < mn$, i.e.,

$$\boxed{k < \frac{mn}{m+n}.}$$

For square $n \times n$: ratio $= 2k/n$, worth when $k < n/2$. □ □